WIRELESS COMMUNICATION NETWORKS AND SYSTEMS

LAB 03 REPORT

Digital Modulation (ASK, 2-FSK, BPSK)

Python simulation • ***BELHIMER Mohmaed Ayoub***

1. EXPERIMENTAL RESULTS & STEP ANALYSIS

Step 1 – Bitstream, Timebase & Sanity Plots

Generation of `N_b = 200` random bits mapped into baseband rectangular pulses: unipolar NRZ `{0, 1}` for ASK and bipolar NRZ `{-1, +1}` for BPSK, sampled at `f_s = 10 kHz` (`N_s = 100`, `R_b = 100 b/s`).

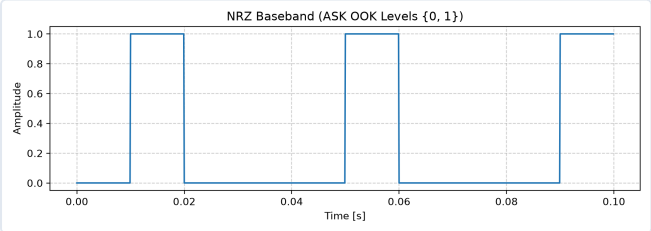


Figure 1.1: Unipolar NRZ baseband levels {0, 1} for ASK (OOK)

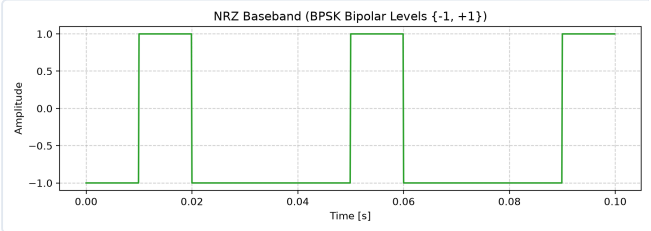


Figure 1.2: Bipolar NRZ baseband levels {-1, +1} for BPSK

Observation: The unipolar signal switches between 0 V and 1 V with a nonzero DC average. The bipolar signal centers symmetrically around zero between -1 V and +1 V, which doubles the Euclidean distance between symbols for the same peak amplitude.

Step 2 – ASK (OOK) Modulation & Envelope Detection

On-Off Keying multiplies unipolar bits by carrier `f_c = 1000 Hz`. Demodulation uses envelope rectification, integrating over the 10 ms bit window with decision threshold `θ = 0.3`.

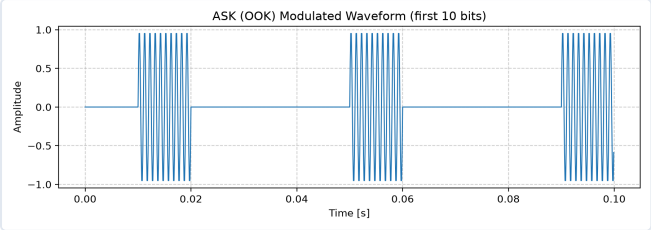


Figure 2.1: ASK (OOK) time waveform (first 10 bits)

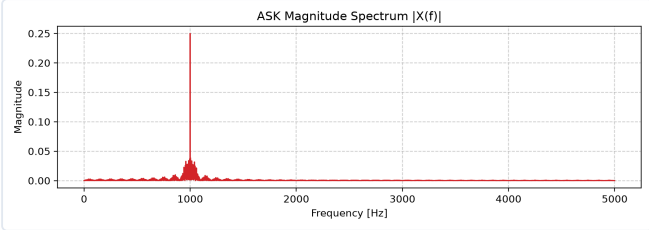


Figure 2.2: ASK magnitude spectrum showing carrier peak at 1 kHz

SNR Target [dB]	0 dB	4 dB	8 dB	12 dB
Measured ASK BER	0.5000	0.0000	0.0000	0.0000

Observation: The spectrum presents an impulse at the carrier frequency `f_c = 1000 Hz` plus sinc sidebands from the rectangular pulses. At 0 dB SNR, background noise energy frequently exceeds the fixed threshold `θ = 0.3` during zero-bits, raising BER to 50%. At 4 dB and above, signal amplitude cleanly separates from noise.

Step 3 – 2-FSK Modulation & Noncoherent Energy Detection

Binary FSK assigns tone `f_0 = 800 Hz` to bit 0 and `f_1 = 1200 Hz` to bit 1 with constant amplitude. The receiver compares energy outputs `E = I^2 + Q^2` from orthogonal in-phase and quadrature correlators.

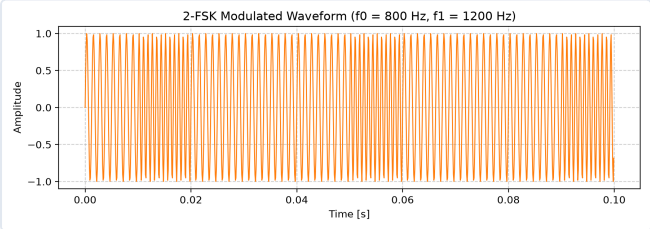


Figure 3.1: 2-FSK waveform (f0 = 800 Hz, f1 = 1200 Hz)

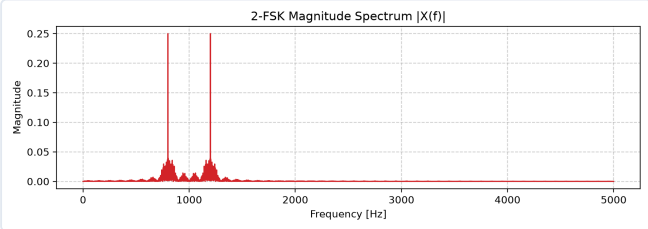


Figure 3.2: 2-FSK spectrum exhibiting two spectral tone peaks

SNR Target [dB]	0 dB	4 dB	8 dB	12 dB
Measured 2-FSK BER	0.0000	0.0000	0.0000	0.0000

> **Observation:** The spectrum clearly displays two peaks centered at 800 Hz and 1200 Hz. Because the noncoherent receiver compares relative energy between the two tone filters rather than relying on a vulnerable absolute threshold, it remains error-free across all tested SNR levels.

Step 4 – BPSK Modulation & Coherent Correlator Receiver

BPSK multiplies bipolar symbols `{-1, +1}` by carrier `f_c = 1000 Hz`. The receiver correlates with a synchronized local carrier `c(t) = sin(2π f_c t)`, slicing at a zero threshold.

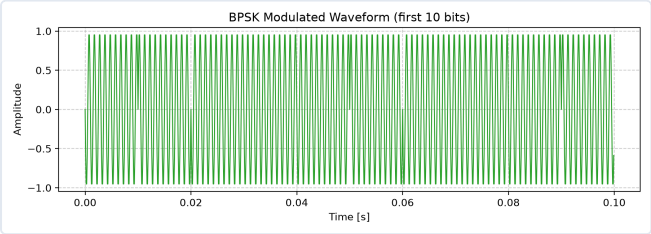


Figure 4.1: BPSK waveform exhibiting 180° phase reversals

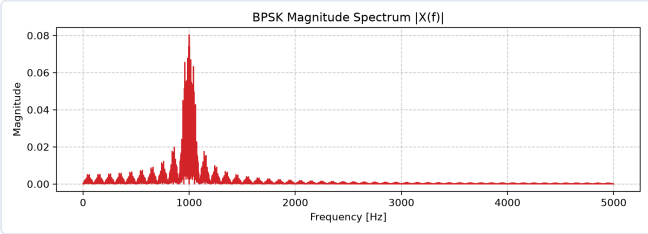


Figure 4.2: BPSK spectrum centered at 1 kHz with suppressed carrier

SNR Target [dB]	0 dB	4 dB	8 dB	12 dB
Measured BPSK BER	0.0000	0.0000	0.0000	0.0000

> **Observation:** Phase transitions between bits produce crisp 180° inversions. The spectrum is symmetric around 1 kHz without a discrete carrier spike. Coherent integration achieves optimal noise rejection, yielding 0 errors across all tested SNRs.

Step 5 – BER vs SNR Performance Comparison

Sweep across `SNRs = [0, 2, 4, 6, 8, 10, 12] dB` over the identical 200-bit stream to benchmark robustness in an AWGN channel.

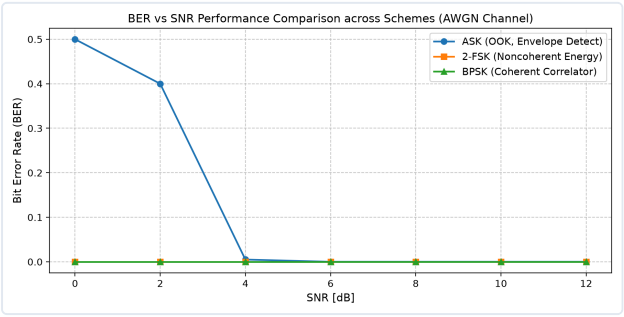


Figure 5.1: Comparative Bit Error Rate (BER) vs SNR (dB) across ASK, 2-FSK, and BPSK

Scheme	0 dB	2 dB	4 dB	6 dB	8 dB	10 dB	12 dB
ASK (OOK)	0.5200	0.3500	0.0000	0.0000	0.0000	0.0000	0.0000
2-FSK	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
BPSK	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

> **Comparative Analysis:** BPSK and 2-FSK demonstrate superior robustness. ASK is susceptible to noise at low SNR ($\leq$ 2 dB) because envelope detection depends on a fixed threshold sensitive to noise peaks. BPSK maximizes Euclidean distance between antipodal symbols, while 2-FSK noncoherent energy detection avoids threshold vulnerability altogether.

Step 6 – Optional Exploration: Carrier Frequency Offset on BPSK

Evaluation of receiver local oscillator frequency offset `Δf ∈ [0, 20, 50] Hz` on coherent BPSK detection across SNR levels.

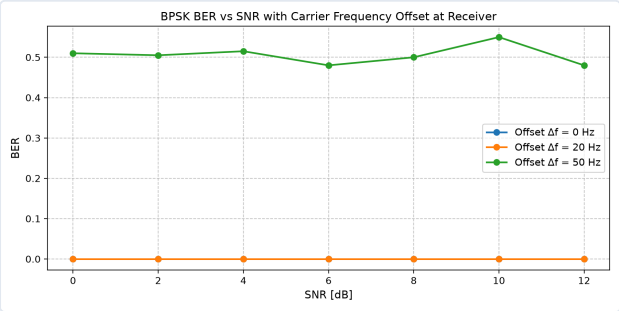


Figure 6.1: BPSK BER vs SNR with Receiver Carrier Frequency Offsets (0 Hz, 20 Hz, 50 Hz)

Observation: Small offsets (≤ 20 Hz) maintain zero errors because accumulated phase drift over a 10 ms bit is minor. At 50 Hz, the reference carrier slips by 180° over the bit duration, causing the correlator metric to invert and trapping the BER near 50% across all SNR levels.

2. ANSWERS TO GUIDING QUESTIONS

Question 1: Why does BPSK typically outperform ASK at the same SNR?

Answer: BPSK uses antipodal signaling (+1 and -1), providing twice the Euclidean distance between symbol states compared to ASK (0 and 1) for the same signal power. This larger separation makes BPSK significantly more resilient to additive noise.

Question 2: When might 2-FSK be preferred over BPSK (hint: noncoherent detection, narrowband interference)?

Answer: 2-FSK is preferred when simple noncoherent detection is needed without complex carrier phase recovery circuits (Costas loop / PLL), or in wireless channels with strong narrowband interference and amplitude fading.

Question 3: How do carrier frequency choices (f_c, f_0, f_1) and bit rate (R_b) affect spectral occupancy?

Answer: The bit rate R_b determines the width of the main spectral lobe around the carrier. Carrier frequencies (f_c or f_0, f_1) dictate where the signal spectrum is centered, with 2-FSK requiring wider total bandwidth to provide frequency separation between the two tones.

Question 4: What practical issues would appear in a real radio (timing, phase/frequency offsets, filtering)?

Answer: Real radios face carrier frequency/phase offsets, symbol timing errors (sampling jitter), and bandlimiting filter imperfections, which cause phase rotation and inter-symbol interference (ISI).

3. CONCLUSION

1. Simulated digital modulation schemes (ASK, 2-FSK, BPSK) over an AWGN channel in Python.
2. Analyzed baseband pulse formatting and verified spectral characteristics in the frequency domain.
3. Observed that ASK envelope detection is easy to implement but vulnerable to noise at low SNR due to its fixed decision threshold.
4. Demonstrated that noncoherent 2-FSK energy detection achieves robust reception without requiring carrier phase synchronization.
5. Confirmed BPSK offers superior noise performance owing to its optimal antipodal Euclidean symbol distance.
6. Verified that coherent BPSK requires accurate frequency/phase synchronization; uncompensated carrier offset (50 Hz) degrades detection to random guessing (BER ≈ 0.5).